PORTFOLIO CASE STUDY

SQL-Based Audit Transaction Analysis

Using MySQL to review transaction activity, approval status, department spending, and vendor risk

PROJECT DETAIL	DESCRIPTION
Primary tool	MySQL / MySQL Workbench
Database	audit_practice
Core tables	transactions; vendors
Domain	Audit analytics, transaction monitoring, vendor risk
Data type	Illustrative practice dataset

Project outcome A commented SQL query pack that summarizes transactions, checks data quality, analyses departments and vendors, and identifies records that warrant further review.

Important scope note: This project demonstrates analytical methods on practice data. High-risk classifications, pending statuses, large amounts, or matching records are screening indicators—not proof of fraud, loss, or control failure.

1. Executive Summary

This project applies MySQL to an audit-oriented transaction dataset to show how structured queries can help an auditor organize transaction activity, compare department spending, examine approval status, and assess transaction exposure associated with vendors classified by risk rating. The analysis uses the existing audit_practice database and its transactions and vendors tables.

The query pack progresses from baseline summaries and data-quality checks to aggregations, conditional logic, joins, CTEs, and a window function. It is designed to be reproducible in MySQL Workbench and to produce evidence that can be reviewed alongside source documentation and applicable company policy.

2. Business Problem and Objectives

Audit teams may need to review many transaction records to understand spending patterns, locate pending or incomplete approval information, and identify transactions associated with vendors classified as higher risk. Manually reviewing records without an initial analytical summary can make it harder to prioritize testing efficiently.

The project objectives are to:

- Summarize total transaction activity and key descriptive statistics.
- Compare transaction value and transaction count across departments.
- Check selected fields for missing values and make missing approval statuses visible.
- Join transactions to vendor reference data to analyze risk classifications.
- Identify high-value transactions and pending transactions associated with high-risk vendors.
- Demonstrate more advanced SQL approaches, including conditional aggregation, a common table expression (CTE), and ROW_NUMBER().
- Provide a framework for potential duplicate-payment screening when the required payment-method field is available.

3. Data and Model

3.1 Tables and fields

Table	Fields used	Role in the analysis
transactions	transaction_id, department, amount, approval_status, vendor_id	Fact-like table holding transaction-level records.
vendors	vendor_id, vendor_name, category, risk_rating	Reference table adding vendor name, category, and risk classification.

The project joins transactions to vendors using vendor_id. A transaction can be matched to its vendor record to expose vendor name and risk rating alongside transaction amount and approval status. The SQL script starts with SHOW COLUMNS statements so the actual database schema can be verified before running the analysis.

Schema caution Optional screening queries use transaction_date or payment_method only if those columns exist in the local transactions table. They remain commented out in the script until verified.

4. Tools and SQL Techniques

Technique	How it is applied
SELECT, WHERE, ORDER BY	Retrieve and prioritize records for review.
COUNT, SUM, AVG, MIN, MAX	Describe transaction volume and monetary values.
GROUP BY and HAVING	Compare departments and filter aggregate results.

CASE WHEN and COALESCE	Create amount bands and display missing approval status clearly.
INNER JOIN and LEFT JOIN	Combine transactions with vendor reference information, including unmatched records where relevant.
Conditional aggregation	Calculate counts and amounts for pending and high-risk categories in one grouped result.
CTE	Make the high-risk pending population easier to read and summarize.
ROW_NUMBER() window function	Rank transactions by amount within each department.
GROUP_CONCAT and duplicate screening	List relevant transaction identifiers; optional duplicate screening groups by vendor, amount, and payment method when available.

5. Analysis and Findings

The figures below are based on results established during earlier exercises in the user's practice database. They should be re-run against the current database before the project is published, especially if source records have changed.

Analysis	Observed result	Audit interpretation
Department spending	IT: ₹1,090,000; Finance: ₹358,000; Operations: ₹331,000; Sales: ₹278,500; HR: ₹77,500.	IT has the highest recorded transaction value in this practice dataset; this is a prioritization signal, not a conclusion about control effectiveness.
Department threshold	HAVING SUM(amount) > ₹500,000 returned IT only.	Demonstrates filtering groups based on an aggregate threshold.
Transaction value bands	High: 6; Medium: 9; Low: 16. High > ₹100,000; Medium ₹50,000–₹100,000; Low otherwise.	These are amount-based bands, not the vendors' separate risk_rating classifications.
High-risk vendor-linked transactions	IDs 1003, 1008, 1012, 1015, 1019, 1025, 1029, 1031; combined transaction value ₹1,145,000.	Records associated with vendors classified High can be selected for further documentation and authorization checks.
High-risk transactions over ₹100,000	IDs 1008, 1015, 1019, 1025, 1029.	High value and high vendor risk together can guide audit prioritization; the combination does not itself show an exception.
Pending + high-risk vendor	IDs 1015 and 1025.	Prioritize checking approval trail, invoice/PO evidence, service or goods receipt evidence, and why the status remains Pending.
Data-quality checks	Earlier checks returned 0 missing vendor_id and 0 missing amount values. Transaction 1031 was identified with a NULL approval_status.	Validate source fields and determine whether the missing approval status is expected, a data issue, or a control exception under policy.

5.1 Department amounts

Department	Total transaction amount
IT	₹1,090,000
Finance	₹358,000
Operations	₹331,000
Sales	₹278,500
HR	₹77,500
Total across departments	₹2,135,000

5.2 Vendor risk is not the same as transaction value

The query pack intentionally keeps two concepts separate: (1) transaction value bands created from the amount field and (2) vendor risk ratings taken from the vendors table. A transaction above ₹100,000 is high

value under the chosen analytical rule, but that does not automatically make its vendor high risk. Similarly, a High vendor rating is a classification that should be supported by the organization's vendor-risk criteria.

6. Audit Observations and Recommended Procedures

Area for review	Recommended procedures	What the procedures establish
Pending transactions linked to high-risk vendors: 1015 and 1025	Inspect invoice details, purchase order or contract, goods receipt/service-completion evidence, approval letter and date, payment workflow history, and documented reason for pending status.	Whether transaction support and approval comply with the applicable process and whether further follow-up is needed.
High-risk vendor activity	Review vendor onboarding and risk-classification support, historical transactions, contracts, vendor master details, bank-detail changes and approvals, and payment authorization evidence.	Whether the vendor classification and payment records are appropriately supported and controlled.
Missing approval status on transaction 1031	Trace the record to the source system and supporting documentation; determine whether approval was obtained, is still pending, or was not captured correctly.	Whether the NULL represents a data-quality issue, a legitimate workflow state, or a potential control exception.
Potential duplicate payment indicators (conditional on payment_method availability)	Group records by vendor_id, amount, and payment_method; inspect possible matches, then compare invoice number, transaction date, supporting documents, and settlement evidence.	Whether similar records are genuinely duplicates or separate legitimate transactions.

Policy dependency Any monetary approval threshold (for example, ₹100,000) must come from the organization's policy or delegated authority matrix. It should not be treated as a universal requirement.

7. Limitations and Responsible Interpretation

- The database is an educational practice dataset, not a verified representation of a real company.
- Risk ratings are accepted from the vendor reference table; this project does not independently validate the criteria used to assign them.
- An amount threshold is an analytical rule selected for screening and may not match policy thresholds.
- Potential duplicate matches require supporting-document review; matching fields alone are not enough to confirm duplication.
- The project has no employee-level analysis unless an employee identifier is available in the schema.
- SQL outputs are leads for audit testing. Any conclusion about control failure or loss requires corroborating evidence and applicable criteria.

8. Conclusion

This project demonstrates how MySQL can convert transaction-level data into structured audit-review information. Aggregation and grouping support spend comparisons; CASE and COALESCE improve classification and visibility of missing approvals; joins bring vendor risk information into transaction analysis; and CTEs and window functions demonstrate more advanced query design. The output provides a reproducible starting point for selecting transactions for further testing rather than replacing professional audit judgment or source-document verification.

9. Portfolio Summary for CV

SQL-Based Audit Transaction Analysis | MySQL Analyzed transaction data using MySQL to summarize department spending, review approval statuses and missing values, join transactions with vendor risk information, and identify high-risk pending records for further investigation. Applied aggregation, CASE WHEN, COALESCE, joins, conditional aggregation, CTEs, and window functions. Used an illustrative practice database and treated screening results as audit leads requiring corroboration.

Appendix: Project files and execution

The accompanying SQL script, SQL_Audit_Transaction_Analysis.sql, contains commented queries organized by analysis objective. It is read-only and does not insert, update, or delete data. Run it in MySQL Workbench against audit_practice, verify available columns first, and save result screenshots only after confirming the displayed outputs match the current database.

Suggested evidence to capture for a portfolio: (1) department spending query and result, (2) high-risk vendor transactions, (3) pending high-risk result, (4) data-quality check, and (5) one advanced query using a CTE or ROW_NUMBER(). Avoid displaying private company data or real personal/vendor banking information.